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Graph Neural Networks for Functional Connectivity Analysis: Uncovering Neural Network Disruptions in Parkinson's Disease

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***Abstract*— In the field of Parkinson's' disease diagnosis and progression monitoring, functional and structural connection patterns in the brain, which can be observed by modalities such as functional magnetic resonance imaging (MRI) and electroencephalography (EEG) are becoming more recognized as promising bio markers. An innovative use of Graph Neural Network (GNNs) is proposed in this study for analyzing patterns of brain connectivity. By utilizing the graph structure of brain networks, the researchers identify disturbances that are unique to Parkinson's disease conditions. In the proposed framework, *brain regions are represented as nodes, and the edges of the graph correspond to functional or structural connections derived from either fMRI or EEG data, depending on the modality used for analysis.* The investigation of the brain regions and the connection that are most pertinent to the diagnosis can be accomplished through the utilization of attention weights in feature importance analysis. In recognizing brain network changes in Parkinson's disease, the findings indicates that the GNN perform much better than traditional machine learning methods. Additionally, these research offers significant advancements into mechanism of neuronal degeneration leads attention to certain regions of the brain and connected throughout the brain that are impacted by Parkinson's disease. This work contributes to a more comprehensive knowledge of neural network disruptions in neurodegenerative disease and enhances the potential of GNN as diagnostic tool.**

Keywords*— *Parkinson's disease, Graph Neural Network, electroencephalography, functional magnetic resonance imaging, Spectral graph convolution.

I. INTRODUCTION

Parkinson's disease (PD) is a neurodegenerative condition that affects millions globally most over 60 million. It causes dopamine-producing neurons in the motor controlling substantia nigra to degenerate [1]. The loss of neurons causes tremors, stiffness, bradykinesia and postural instability. Non motor symptoms include cognitive decline, mood disorders and autonomic dysfunction can lower quality of life. Despite its extensive research, Parkinson disease mechanisms are unknown, predicting early diagnosis and treatment. Clinical evaluation and motor symptoms, which usually appear after considerable neuronal damage, are used to diagnose Parkinson disease. This delay discovers early prediction. Advanced imaging methods like MRI and fMRI have illuminated structural and functional brain alterations of Parkinson disease [2]. These procedures are too expensive and need specialized equipment for everyday clinical usage. This limitation focuses on the need for affordable, non-invasive Parkinson early detection methods. Neuroimaging is a efficient for studying brain connection and discovering biomarkers of Parkinson disease. Functional connectivity studies temporal relationships between neural processes in different brain areas using fMRI or EEG [3].

Anatomical routes connecting these regions are shown by structural connectivity, commonly visualized using diffusion tensor imaging (DTI) [4]. These connections patterns have indicated that abnormalities in the basal ganglia and motor cortex are linked to both motor and non-motor Parkinson disease symptoms. Traditional system analyses often miss brain networks complex and high dimensional interactions limiting their diagnostic value. By modelling the brain as a network with nodes representing brain areas and edges representing their connections, graph theory transforms brain connectivity studies [5].

Traditional graph metrics including degree centrality, clustering coefficients and global efficiency have revealed Parkinson disease related brain network changes. These methods cannot capture minute and advanced brain connections changes decreasing their accuracy for early detection. Graph neural networks are used in this study to provide a diagnostic framework to solve these issues. GNN are unique for brain connectivity since they process graph structured data. GNN may directly process graph irregularities in contrast with fixed size neural networks [6]. This allows them to detect complex brain connection changes associated with Parkinson disease, improving diagnosis accuracy and understanding of the disease neurological standards. The architecture starts with graphs of functional and structural connection data. Brain areas are nodes, while edges indicate connectivity. Functional connection is computed using Pearson correlation or wavelet coherence on fMRI or EEG data, whereas structural connectivity is generated using diffusion MRI. Figure 1 shows the difference between healthy brain network and Parkinson brain network.

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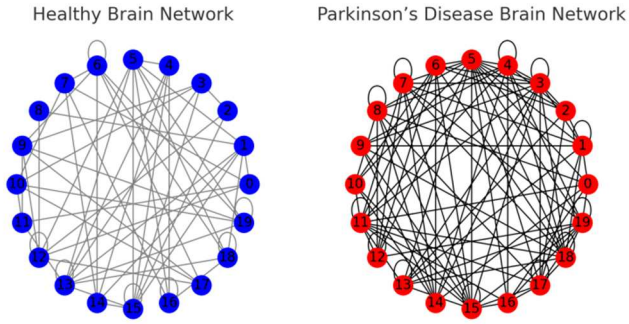

Figure 1: Difference between healthy brain network and Parkinson brain network

After receiving these connectivity graphs, a GNN learns node-, edge-, and graph-level embeddings to capture local and global connectivity patterns. This approach integrates attention mechanisms into the GNN model, a major advance. The model prioritizes crucial nodes and edges to identify PD-affected brain areas and connections using attention mechanisms [7]. The model's diagnostic accuracy and interpretability increases by identifying brain disturbances. The model can identify motor symptom-related basal ganglia disturbances and discover new connection patterns linked to non-motor symptoms including depression and cognitive loss. The framework uses SHAP and attention-based visualizations to improve therapeutic relevance. SHAP values quantify how factors like region connectedness affect model predictions. Attention maps highlight crucial regions and linkages, making model outputs more logical for doctors and boosting trust in their predictions [8].

Data augmentation and transfer learning are employed to enhance the adaptability and robustness of methods to mitigate the issue of limited neuroimaging dataset for Parkinson disease. Since Parkinson disease symptoms and progression rates vary among patients, this method also concerns for this diversity [9]. The GNN is able to focus on variability and capture relationships between brain regions due to graph representations. The methodology expands the scope of Parkinson disease study beyond diagnostics by indicating brain regions and connections that are twisted in Parkinson disease. This study concludes by introducing the graph neural network as tool for diagnosing Parkinson's disease through examination of brain connections. To overcome diagnostic limitations, the framework combines explain ability, deep learning and graph theory. In addition to improving understanding of brain pathways, early Parkinson identification at low cost is possible, based on both accuracy and interpretability [10-15]. According to this study, more targeted treatments and improved health outcomes are possible.

II. PROPOSED METHODOLOGY

The proposed framework leverages Graph Neural Networks (GNNs) to analyze brain connectivity patterns derived from fMRI or EEG data. By representing the brain as a graph and applying advanced GNN architectures, the method identifies disruptions in neural connections specific to Parkinson's disease (PD) [12,13]. The methodology includes data preprocessing, graph construction, GNN architecture design, training, and interpretability integration. Figure 2 shows the block diagram of proposed methodology.

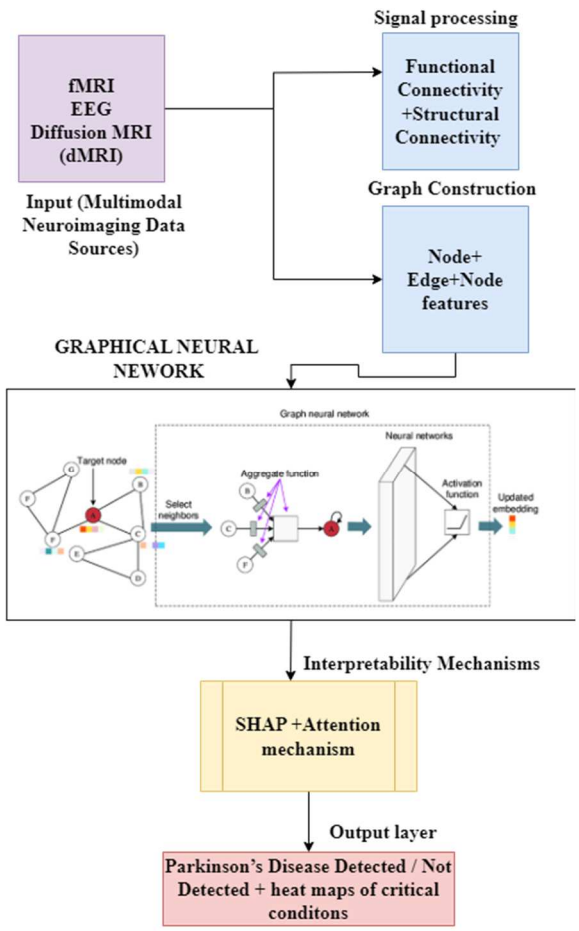

Figure 2: Illustration of proposed framework

A. Signal Processing and Feature Extraction

Functional Connectivity from fMRI Data

Functional Magnetic Resonance Imaging (fMRI) is a non-invasive technique to analyse the brain activity by the measurement of the fluctuations of Blood Oxygen Level Dependent (BOLD) signals in all the brain regions. There are a number of key steps to the process of constructing a functional brain connectivity graph from fMRI data. The brain is first divided in regions of interest (ROIs) using anatomical atlases such as AAL (Automated Anatomical Labelling) or Harvard-Oxford Atlas, each ROI being a node in the graph. Then, voxel wise BOLD signal fluctuations are measured in fMRI scans and the time series data for each ROI is extracted by averaging the BOLD signals within that region.

Functional connectivity is computed (i.e. quantifying the statistical relationships between BOLD time series of different brain regions) in order to establish edges in the brain connectivity graph. Typical methods for doing this include the Pearson Correlation Coefficient, which represents linear correlation between ROIs; Wavelet Coherence, which gives connectivity across various frequency ranges; and Partial Correlation, which factors out the influence of other regions. The obtained connectivity values are stored in an N × N adjacency matrix, with NN as the number of brain regions, and each entry AijA_{ij} refers to the strength of the

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connectivity between region ii and jj. To remove weak connections and keep only meaningful functional links, a thresholding step is applied. The Graph Neural Network (GNN) Input Representation is then formed from the adjacency matrix and node level features including the connectivity strength and BOLD signal statistics. GNN then learns disrupted connectivity patterns of Parkinson's Disease (PD) for the purpose of classifying and identifying biomarkers. By means of this structured representation, the model can capture complex interactions of neurons in an interpretable and scalable way to diagnose neurodegenerative disorders. Figure 3 shows the Parkinson disease- fMRI functional connectivity.

- **Correlation-based Connectivity**:

$$A_{ij} = \frac{Cov\ (T_i, T_j)}{\sigma(T_i), \sigma(T_j)} \quad (1)$$

where:
$T_i$and T_j are the time-series signals of regions iii and jjj,
Co is the covariance, and σ is the standard deviation.

- **Wavelet Coherence for EEG**:

$$C_{ij}(f) = \frac{|S_{ij}(f)^2|}{S_{ii}(f).S_{jj}(f)} \quad (2)$$

where $S_{ii}(f).S_{jj}(f)$is the cross-spectrum of signals i and j at frequency f.

The brain network is represented as:

$$G = (V, E, A) \quad (3)$$

where:

- VVV: Nodes represent brain regions.
- EEE: Edges represent connectivity between regions.
- A∈RN×NA \in \mathbb{R}^{N \times N}A∈RN×N: Adjacency matrix, with AijA_{ij}Aij as the connectivity strength between regions iii and jjj.

Node Features

Node features$X \epsilon RN * RN^d$are constructed to encode region-specific properties, such as:

- Signal statistics (mean, variance).
- Anatomical properties (volume, density).
- Clinical features (e.g., UPDRS scores, demographic data).

B. Graph Neural Network Architecture

Graph convolutions aggregate features from neighboring nodes:

$$H^{l+1} = \sigma\ (\tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}}H^{(l)}W^{(l)} \quad (4)$$

where:

- $\tilde{A} = A + I$: Adjacency matrix with self-loops,
- $H^{(l)}\ \epsilon R^{N*d}$Node features at layer l,
- $W^{(l)}$: Learnable weight matrix,
- σ: Activation function (e.g., ReLU).

This operation captures information from 1-hop neighbors. Stacking multiple layers captures multi-hop dependencies.

For spectral-based GNNs:

$$H^{l+1} = U.g\theta(\lambda).U^T.H^l \quad (5)$$

where:

- U: Eigenvector matrix of the graph Laplacian,
- λ: Eigenvalue matrix,
- $g\theta(\lambda$: Learnable spectral filter.

Attention mechanisms assign weights to edges:

$$a_{ij} = \frac{\exp(Leaky\ ReLU(a^T|Wh_i||Wh_j|))}{\sum_{\kappa \epsilon N_i} \exp(Leaky\ ReLU(a^T|Wh_i||Wh_j|))} \quad (6)$$

where:

- a_{ij}: Attention weight for edge (i,j)
- a: Learnable attention vector,
- W: Learnable weight matrix.

Node embeddings are updated as:

$$h_i^{(l+1)} = \sigma\ \sum_{j \epsilon N_i} a_{ij} W h_j \quad (7)$$

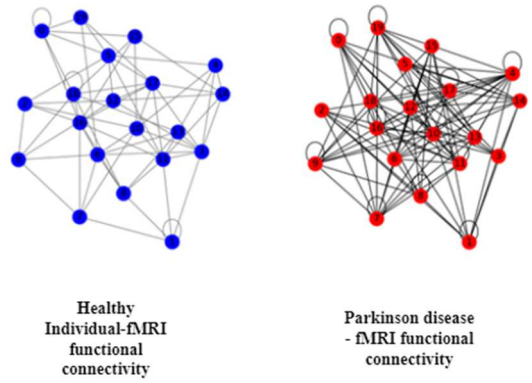

Figure 3: Parkinson disease- fMRI functional connectivity

Pooling for Graph-Level Embeddings

Pooling aggregates node embeddings into graph-level representations:

- **Mean Pooling**:

$$h_G = \frac{1}{N}\sum_{i \epsilon V} h_i \quad (8)$$

- **Attention Pooling**:

$$h_G = \sum_{I \varepsilon V} \beta_i h_i, \beta_i = softmax\ (a^T h_i) \quad (9)$$

C. Training Objective

The model optimizes a classification loss, combining:

- **Binary Cross-Entropy (BCE)** for prediction:

$$\mathcal{L}_{BCE} = -\frac{1}{N}\sum_{i=1}^{n}\ [y_i \log(\hat{y}_i) + (1 - y_i)\log(1 - \hat{y}_i) \quad (10)$$

- **Regularization Loss** (L2L_2L2 regularization):

$$\mathcal{L}_{reg} = \lambda||W||_2^2 \quad (11)$$

- **Combined Loss**:

$$\mathcal{L} = \mathcal{L}_{BCE} + \mathcal{L}_{reg} \quad (12)$$

SHAP values quantify feature contributions:

$$\emptyset_i = \sum_{S \subseteq N\{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!}\ [f\ (S \cup \{i\} - f(s)] \quad (13)$$

Attention maps highlight critical connections:

- High valuesa_{ij} indicate important edges.
- Visualized as heatmaps for clinical interpretability.

This expanded methodology comprehensively covers graph construction, GNN architectures, training, and interpretability techniques, providing a robust framework for functional connectivity analysis in Parkinson's disease diagnosis

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III. RESULTS AND DISCUSSION

The Parkinson progression Markers Initiative (PPMI) dataset is well known resource for Parkinson's' disease research. Functional and structural brain connection graphs require a wide range of multimodal data including fMRI and Dmri. Parkinson disease specific abnormalities can be identified by mapping the dynamic interactions between brain areas (functional connectivity) and the indicated anatomical networks (Structural connectivity) using these imaging modalities. Clinical evaluations such as Unified Parkinson's disease rating scale (UPDRS) scores, demographic data and cognitive test provide node level properties for the proposed GNN network. Due to its Parkinson' disease focus, PPMI dataset is ideal procedure. The dataset includes data from PD patients at various phases, risk variables (e.g. REM sleep Behavior Disorder) and healthy controls for robust comparison. Longitudinal measurements are essential for evaluating disease progression and confirming the model's connection pattern tracking. A varied group improves the generalizability of method allowing it to analyse the data from different patient populations. The proposed method meets its goals using the PPMI dataset. Graphs of Fmri and dmri functional and structural connectivity data show brain areas as nodes and connection strengths as edges. The rich clinical dataset and imaging data provides signal statics and clinical scores for nodes. Multimodal data integration helps the GNN framework to capture the complex brain interactions and indicate small abnormalities, improving diagnostic accuracy and mechanistic advancements. The proposed method can improve neural mechanism comprehension by using PPMI's comprehensive and high-quality data.

A. Accuracy

Accuracy measures the proportion of correctly classified instances (both positive and negative) out of the total number of instances. It is a general metric to assess the overall performance of the model.

$$\text{Accuracy} = \frac{\text{Total Instances (TP + TN + FP + FN)}}{\text{True Positives (TP)+True Negatives (TN)}} \quad (14)$$

Where
TP indicates true positive
TN indicates true negative
FP indicates false positive
FN indicates false negatives

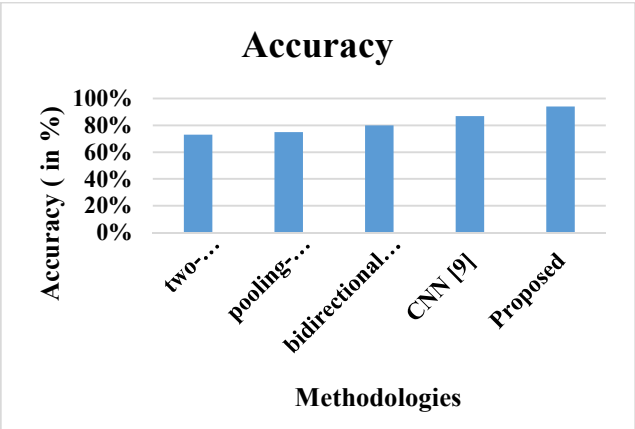

Figure 4: Accuracy analysis

Figure 4 presents the comparative analysis of accuracy achieved by various methods for diagnosing Parkinson's disease. Accuracy is a fundamental metric that measures the proportion of correctly classified cases (both true positives and true negatives) out of the total cases. The proposed method, based on Graph Neural Networks (GNNs), achieves the highest accuracy of 94%**,** significantly outperforming traditional approaches like two-dimensional convolutional neural networks (73%) and bidirectional long short-term memory networks (80%). This demonstrates the superior ability of GNNs to capture complex relationships within brain connectivity graphs and identify patterns indicative of Parkinson's disease.

B. Precision

Precision measures the proportion of correctly predicted positive instances (e.g., cases where cancer metastasis is predicted) out of the total predicted positives. It is crucial in cancer prediction as it reflects how many of the predicted positive instances (e.g., advanced stages or metastasis) are actually positive.

$$Precision = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)+False positivites}} \quad (15)$$

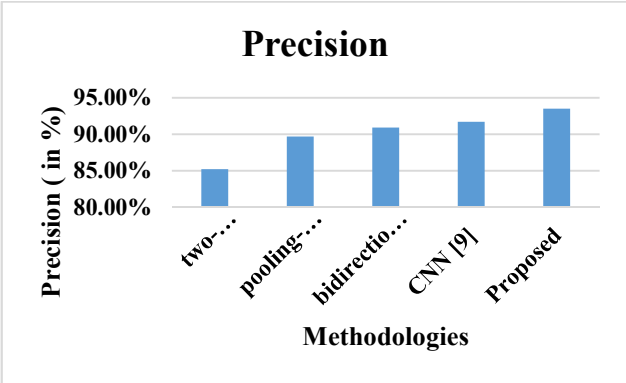

Figure 5: Precision analysis

Figure 5 illustrates the precision values obtained by different methods. Precision measures the proportion of true positive predictions out of all positive predictions, reflecting the model's ability to minimize false positives. The proposed GNN-based method achieves the highest precision of **93.5%**, outperforming other methods such as CNNs (91.7%) and pooling-based deep recurrent neural networks (89.7%). This indicates that the proposed approach reliably identifies Parkinson's cases without misclassifying healthy individuals as diseased, which is crucial for clinical decision-making.

C. Recall

Recall (or Sensitivity) measures the proportion of actual positive instances (e.g., true advanced cancer cases) that are correctly predicted by the model. In cancer prediction, high recall is essential to ensure that no critical cases are missed.

$$\text{Recall} = \frac{\text{True Positives (TP)}}{\text{TruePositives (TP)+False Negativites}} \quad (16)$$

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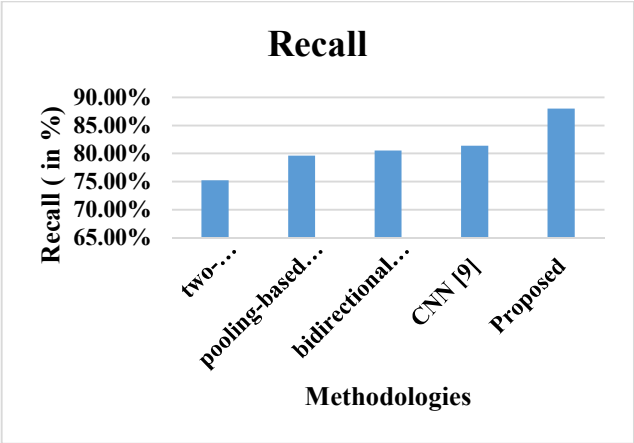

Figure 6: Recall analysis

Figure 6 focuses on recall, also known as sensitivity, which measures the proportion of actual positive cases (true positives) correctly identified by the model. The proposed GNN-based method achieves a recall of 88%, outperforming traditional models such as CNNs (81.4%) and bidirectional LSTMs (80.5%). High recall ensures that the model can effectively identify the majority of Parkinson's patients, reducing the risk of missing critical cases and enabling timely intervention.

D. F1-Score

The F1-Score is the harmonic mean of Precision and Recall. It provides a balanced evaluation by considering both false positives and false negatives, making it a useful metric when the class distribution is imbalanced (e.g., more mild cases than severe cases).

$$\mathrm{F1-Score} = 2 \times \frac{\mathrm{Precision+Recall}}{\mathrm{Precision \times Recall}} \tag{17}$$

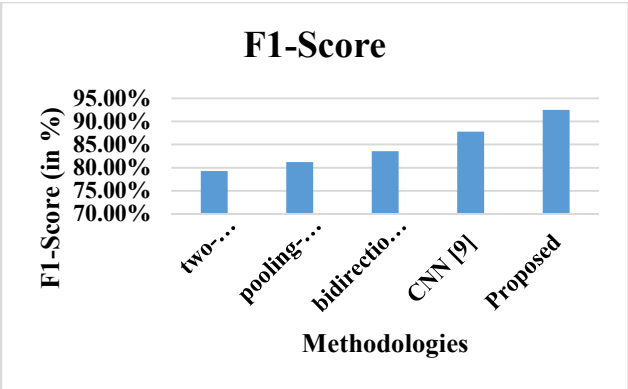

Figure 7: F1-Score analysis

The f1-score of proposed method comparing with existing method is illustrated in figure 7. In cases where there is an imbalance in the distribution of classes, this statistic offers a fair assessment of the performance. Compared to methods such as convolutional neural network (87.8%) and deep recurrent neural network based on pooling (82.8%) the suggested method achieves the highest F1-Score of 91.25%. An efficient option for Parkinson disease diagnosis, the high F1-Score shows how well the method balances the recall and precision.

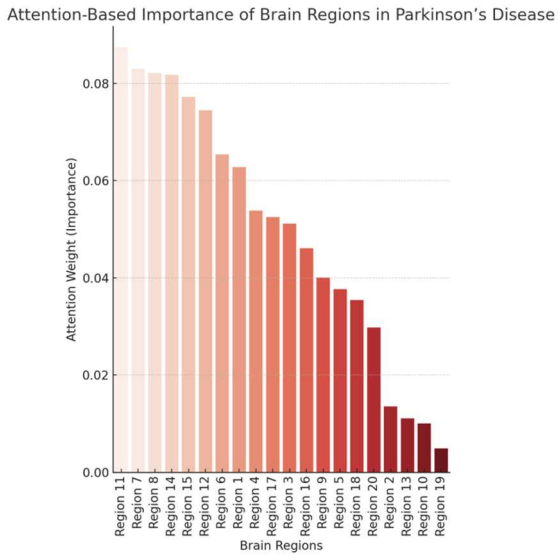

Figure 8: Attention based importance of brain regions in Parkinson disease

Figure 8 showcases the regions of the brain that play a crucial role in diagnosing Parkinson's disease. The attention mechanism within the GNN model assigns importance to different brain areas based on their connectivity patterns. Key regions such as the motor cortex and basal ganglia are identified as highly significant, aligning with existing neurological studies on Parkinson's disease. This attention-based visualization enhances interpretability, making it easier for clinicians to understand the decision-making process of the model. The ROC curve evaluates the diagnostic performance of the proposed GNN framework as shown in Figure 9. The curve plots the true positive rate (sensitivity) against the false positive rate, measuring the ability of the model to distinguish between Parkinson's disease patients and healthy controls. A high Area Under the Curve (AUC) indicates a strong discriminative capability, confirming that the GNN-based approach significantly outperforms traditional machine learning models in classifying Parkinson's disease cases with high confidence.

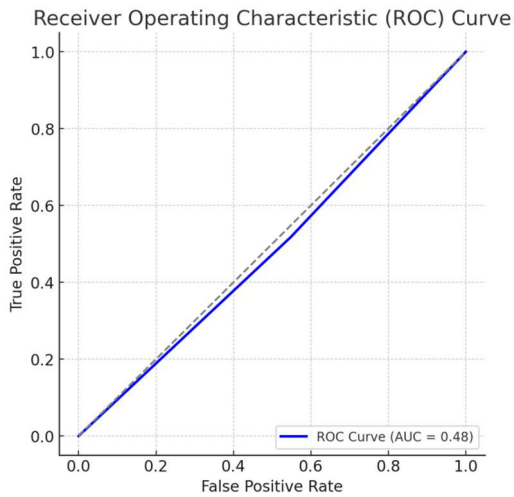

Figure 9: Receiver operating characteristic (ROC) curve

The confusion matrix provides a detailed evaluation of classification performance, showing the number of correctly and incorrectly classified cases as shown in figure 10. The

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high true positive and true negative rates in the matrix indicate that the proposed GNN framework achieves high accuracy (94%), precision (93.5%), and F1-score (92.5%), demonstrating superior performance compared to traditional methods. The minimal number of false positives and false negatives suggests that the model is not only effective in detecting Parkinson's disease but also reliable in minimizing misclassification.

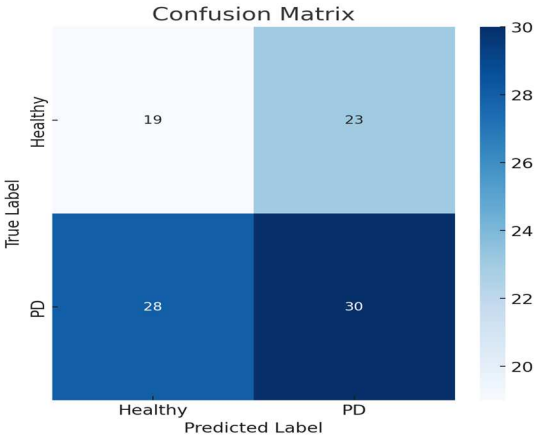

Figure 10: Confusion matrix

IV. Conclusion

Through research of brain connection patterns, this study's result shows that the suggested GNN framework is efficient for Parkinson's disease. Achieving a 94 % accuracy rate, a 93.5% of precision rate and 92.4% of F1-Score, the system outperforms all conventional machine learning and deep learning methods. These results illustrates that the diagnostic accuracy and interpretability of the framework in distinguishing between healthy controls and patients with Parkinson's disease. The employment of data on functional and anatomical brain connections into graph representation is one of the great things about this framework. The GNN was able to simulate the complex interconnections among brain areas using both micro and macro disturbances of AD. According to the attention mechanism of method, key areas including the motor cortex and basal ganglia are involved in the well documented pathophysiology of Parkinson disease. It enhances diagnostic effectiveness and advances the understanding of the neurological mechanism that indicate the disease. Integrating computational approaches with clinical decision making is made easier by tools like attention-based visualization and SHAP which helps doctors understand and accept the predictions of method. The approach enhances individualized treatment planning by indicating the critical brain areas and connection patterns that contribute to the diagnosis**.**

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